

Robotics is an interdisciplinary branch of computer science and engineering. Robotics involves the design, construction, and operation of robots. Control Theory is a subfield of mathematics and engineering that deals with controlling dynamical systems in engineering. PID Controllers (Proportional-Integral-Derivative) are widely used in Industrial Robotics to maintain continuous tasks.

Reinforcement Learning (RL) is a subfield of Machine Learning where an Agent learns to make optimal decisions through trial and error. Q-Learning and Deep Q-Networks (DQN) are foundational algorithms in Reinforcement Learning applied to robotics.

Computer Vision (CV) is a field of artificial intelligence that enables computers to derive meaningful information from visual data. Convolutional Neural Networks (CNNs) are specialized deep learning architectures designed for spatial image recognition. YOLO (You Only Look Once) is a real-time object detection algorithm used in Autonomous Vehicles.

Autonomous Vehicles rely on Computer Vision and Sensor Fusion to navigate environments safely. Sensor Fusion combines data from LiDAR, Radar, and Cameras to produce an accurate environmental spatial map.